BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 026 · Boolean logic, gates and truth tables

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the difference between a stack and a queue when removing items?

Q2. What must be true about an array before you use binary search?

Q3. For int[] numbers = {5, 9, 14, 18, 23}; , what value is stored at zero-based index 3?

Q4. A binary search looks for 23 in the sorted array [2, 7, 11, 16, 23, 31, 44] . Which zero-based indexes are checked, in order?

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I Do · Read new knowledge

Key vocabulary

Recall the term — Boolean

Recall the term — AND

Recall the term — OR

Recall the term — NOT

Recall the term — XOR

Recall the term — logic gate

Recall the term — truth table

New learning

AND — both inputs must be 1 — A gate takes an input (or inputs) and produces one output, Q . For AND, both inputs must be 1. — Example: 1 AND 0 = 0; 1 AND 1 = 1.

OR — at least one input is 1 — Either input can turn Q on. Both can be on too. — Example: 1 OR 0 = 1; 1 OR 1 = 1.

NOT — reverse one input — NOT has just one input . The small circle on the output is the inversion symbol. — Example: NOT 1 = 0; NOT 0 = 1.

XOR (⊻) — exactly one input is 1 — XOR is different from OR: when both inputs are 1, XOR gives 0. Notice the extra curved line at the input. — Symbol: Q = A ⊻ B. Compare: 1 ∨ 1 = 1, but 1 ⊻ 1 = 0.

Words first, then Boolean symbols — These symbols are shorthand for the rules you have just seen: — Say: A OR B .

Connect two gates in order — Say the logic aloud: A OR B, then AND C. — Brackets mean the OR gate is worked out before its output goes into AND.

Check the circuit with a truth table — For three inputs there are 8 combinations . Work out the OR column first, then use that result with C. — CODE:

A	B	C	A∨B	Q
0	0	0	0	0
0	0	1	0	0
0	1	0	1	0
0	1	1	1	1
1	0	0	1	0
1	0	1	1	1
1	1	0	1	0

1 1 1 | 1 | 1

Worked example

Secure door access

1. What switches ON? — CODE:
Q = door unlocks — Start with the result the question is asking us to control.

2. What are the alternatives? — CODE:
A ∨ B — A card OR PIN works. OR (∨) means at least one; both can also be true. Do not write ⊻ here: XOR would wrongly stop access when both card and PIN are valid.

3. What must also be true? — CODE:
(A ∨ B) ∧ C — The building must be open , so join the options with AND.

4. What must be OFF? — CODE:
¬D — Lockdown must not be active. NOT reverses D.

5. Join the chunks — CODE:
Q = (A ∨ B) ∧ C ∧ ¬D — Work backwards: the final AND checks all requirements. Draw OR, AND and NOT first, then connect them to the final AND.

6. Use intermediate columns — For Q = (A ∨ B) ∧ ¬C, three inputs give $2^3 = 8$ combinations. Write every input row, calculate A∨B, then ¬C, then combine those two columns with AND. — CODE:
A B C | A∨B | ¬C | Q
0 0 0 | 0 | 1 | 0
0 0 1 | 0 | 0 | 0
0 1 0 | 1 | 1 | 1
0 1 1 | 1 | 0 | 0
1 0 0 | 1 | 1 | 1
1 0 1 | 1 | 0 | 0
1 1 0 | 1 | 1 | 1
1 1 1 | 1 | 0 | 0 — Example row 010: A∨B = 1 and ¬C = 1; therefore Q = 1. Always complete the intermediate columns before Q.

We Do · Practise with guidance

Successful login

A login succeeds only when the user ID is valid AND the password is correct .

```
A = valid user ID
B = correct password
Q = login succeeds
```

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Find the output, options, requirements and exceptions. Then write the expression and draw the gate.

Which gate?

Are there any alternative routes? What must be true? Write Q, draw the gate and predict the truth-table outputs.

```
A B | Q
0 0 | ?
0 1 | ?
1 0 | ?
1 1 | ?
```

AND: both required

```
Q = A ∧ B
```

Output: login. No OR options or NOT exception. AND joins two requirements.

```
A B | Q
0 0 | 0
0 1 | 0
1 0 | 0
1 1 | 1
```

Two test switches

A test lamp lights when exactly one of switches A and B is ON, but not both .

```
A = switch 1 on
B = switch 2 on
Q = lamp lights
```

Compare this with a different instruction: “at least one switch is on”.

OR or XOR?

Choose the gate and write Q = A ⊻ B for exactly one . How would the symbol and truth table change for at least one ?

```
A B | EXACTLY ONE | AT LEAST ONE
0 0 |      ?      |      ?
0 1 |      ?      |      ?
1 0 |      ?      |      ?
1 1 |      ?      |      ?
```

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Burglar alarm circuit
A burglar alarm has four inputs. Each input is 1 when the condition is true and 0 otherwise:
A: door sensor detects movement
B: motion sensor detects movement
C: alarm is armed
D: test mode is enabled
The alarm sounds (Q = 1) only when at least one sensor detects movement , the alarm is armed , and test mode is off .
Your tasks: (1) Write the Boolean expression for Q. (2) Draw the logic-gate circuit using AND, OR and NOT gates. (3) Label A, B, C, D and Q.

Question 2

Expression and truth table
For P = ¬(A ∨ B) ⊻ C, complete all eight output values.

Question 3

Create logic from words
A cooling fan runs when temperature is high OR load is high, but only when maintenance mode is NOT enabled. Write the expression and truth table.

Question 4

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Expression to gate plan

Convert Q = (A ⊻ B) ∧ ¬C into a gate-by-gate plan and complete its truth table.

Exit Ticket · Answer before checking

A three-input Boolean problem has eight possible input combinations in its truth table.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A stack removes the most recently added item (LIFO); a queue removes the earliest added item (FIFO).

Q2. The values must be sorted in the order expected by the search.

Q3. 18. C# arrays start at index 0.

Q4. Indexes 3 → 5 → 4 (values 16 → 31 → 23). The search narrows the sorted range each time.

You Do pages · Independent questions

Q1. CODE:
A = door sensor; B = motion sensor
C = alarm armed; D = test mode

Q = (A ∨ B) ∧ C ∧ ¬D

1. OR A and B.
2. NOT D.
3. AND (A∨B) with C.
4. AND that result with ¬D to give Q.
Check: A=1, B=0, C=1, D=0 gives Q=1. Changing D to 1 makes Q=0. A three-input final AND is also correct.

Q2. P = ¬(A ∨ B) ⊻ C. Calculate OR, then NOT, then XOR. XOR is 1 when its two inputs differ.
CODE:
A B C | A∨B | ¬(A∨B) | P
0 0 0 | 0 | 1 | 1
0 0 1 | 0 | 1 | 0
0 1 0 | 1 | 0 | 0
0 1 1 | 1 | 0 | 1
1 0 0 | 1 | 0 | 0
1 0 1 | 1 | 0 | 1
1 1 0 | 1 | 0 | 0

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1 1 1 | 1 | 0 | 1
Check: For 000, NOT(0 OR 0)=1 and 1 XOR 0 = 1.

Q3. T = temperature high; L = load high; M = maintenance mode on.
CODE:
F = (T ∨ L) ∧ ¬M

T L M | T∨L | ¬M | F
0 0 0 | 0 | 1 | 0
0 0 1 | 0 | 0 | 0
0 1 0 | 1 | 1 | 1
0 1 1 | 1 | 0 | 0
1 0 0 | 1 | 1 | 1
1 0 1 | 1 | 0 | 0
1 1 0 | 1 | 1 | 1
1 1 1 | 1 | 0 | 0
Check: At least one trigger must be high, and maintenance must be off.

Q4. CODE:
Q = (A ⊻ B) ∧ ¬C

1. XOR A and B.
2. NOT C.
3. AND those outputs.

A B C | A ⊻ B | ¬C | Q
0 0 0 | 0 | 1 | 0
0 0 1 | 0 | 0 | 0
0 1 0 | 1 | 1 | 1
0 1 1 | 1 | 0 | 0
1 0 0 | 1 | 1 | 1
1 0 1 | 1 | 0 | 0
1 1 0 | 0 | 1 | 0
1 1 1 | 0 | 0 | 0
Check: Q=1 only when A and B differ and C=0.

Exit Ticket · Model answer

True. Each of the three Boolean inputs has two states, giving 2 × 2 × 2 = 8 combinations.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: